

Autonomous Data Ferrying From a Self- Organizing/Healing WSN in Disconnected Zones

25-26J-010



PP2 Comments & Actions Taken

PP2 Comment	Action Taken
Need proper demonstration and analysis for the final	Prepared live demo with MS nodes, CH election, data transfer, dashboard update, and test result analysis

Problem & Overall Solution

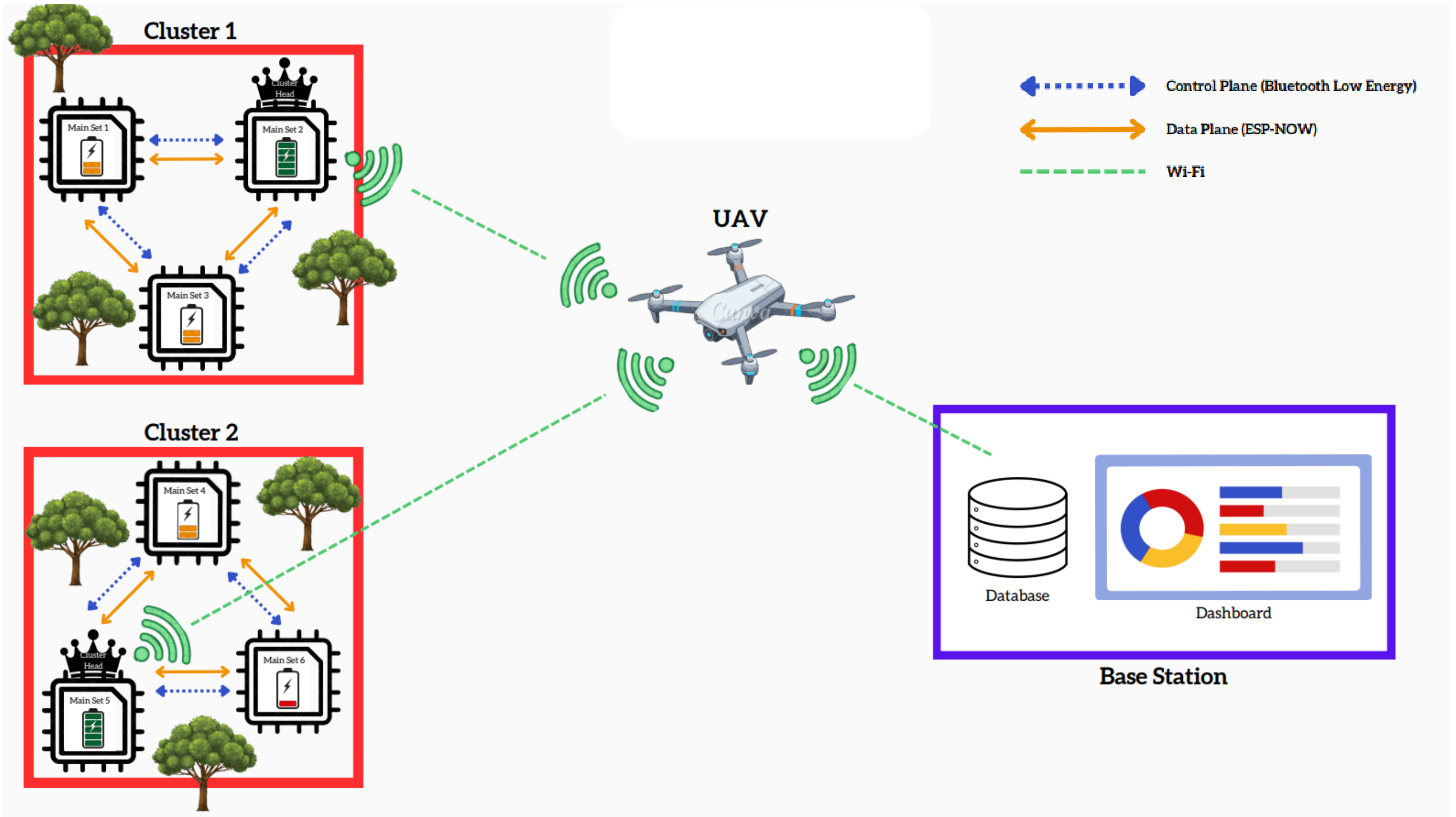
Remote areas such as disaster zones, wildlife areas, and off-grid farms do not always have internet, gateways, or stable power. Traditional WSNs depend on fixed gateways or multi-hop routing, which can drain batteries and fail in disconnected environments.

Our Solution: SKY-MULE

A UAV-assisted self-organizing WSN where sensor nodes collect data, elect a Cluster Head, store data locally, and a UAV visits the CH to collect buffered data.

Design Excellence

- No fixed gateway needed
- Self-organizing CH election
- Solar-powered ESP32-S3 sensor nodes
- TDMA-based ESP-NOW communication
- Persistent SPIFFS storage
- UAV-based data ferrying using ROS 2 and ArduPilot
- Dashboard for monitoring collected data



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MS Hardware & Power Design



Individual Component

Autonomous Multi-Sensor Environmental Monitoring Node

Key Features

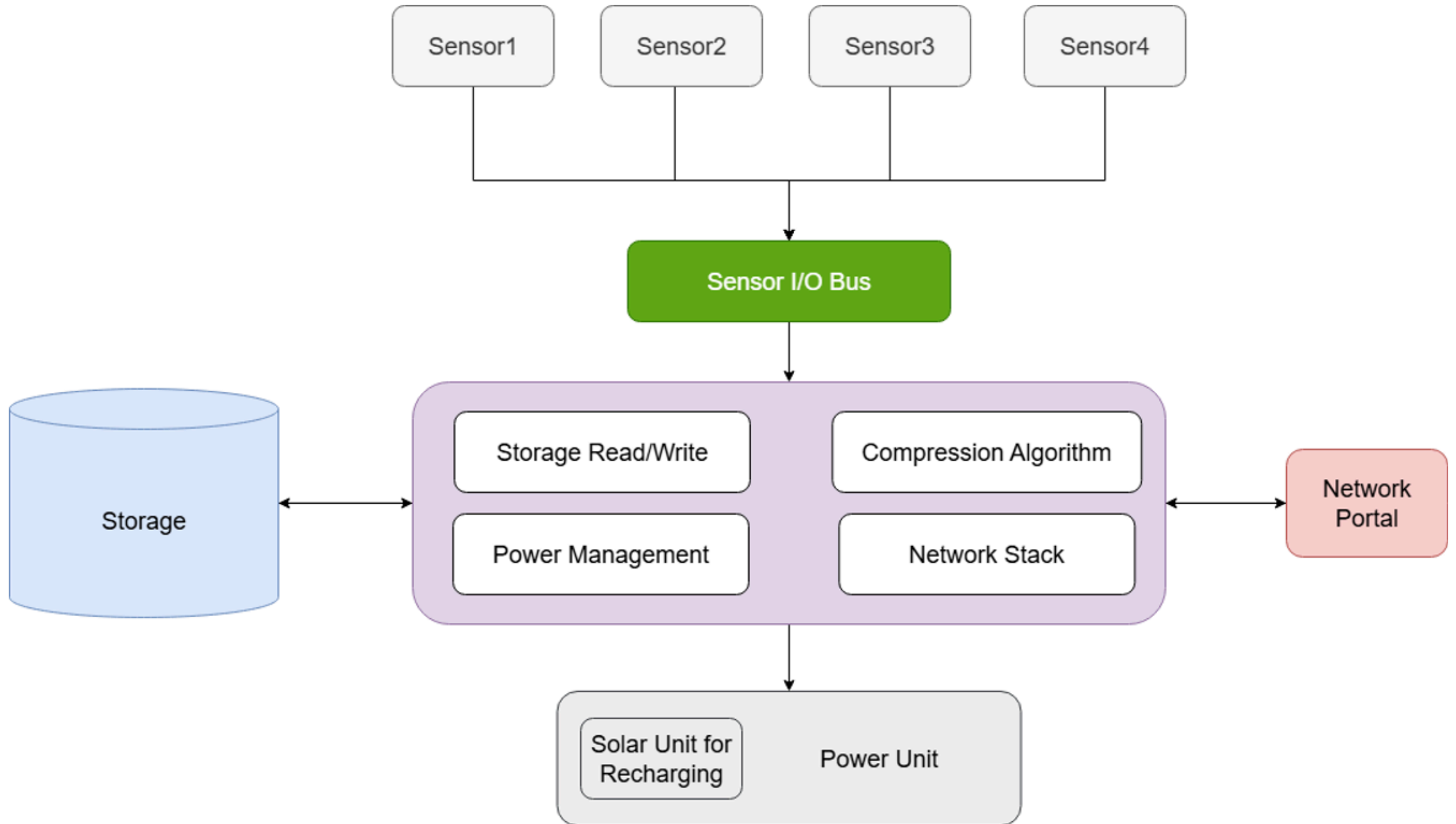
- ESP32-S3-based sensor node
- Sensors: ENS160, INMP441, HMC5883L, BME280
- Three power modes: Normal, Passive, Critical
- Solar charging using CN3065
- Power monitoring using INA219
- Compression comparison: Huffman vs miniz

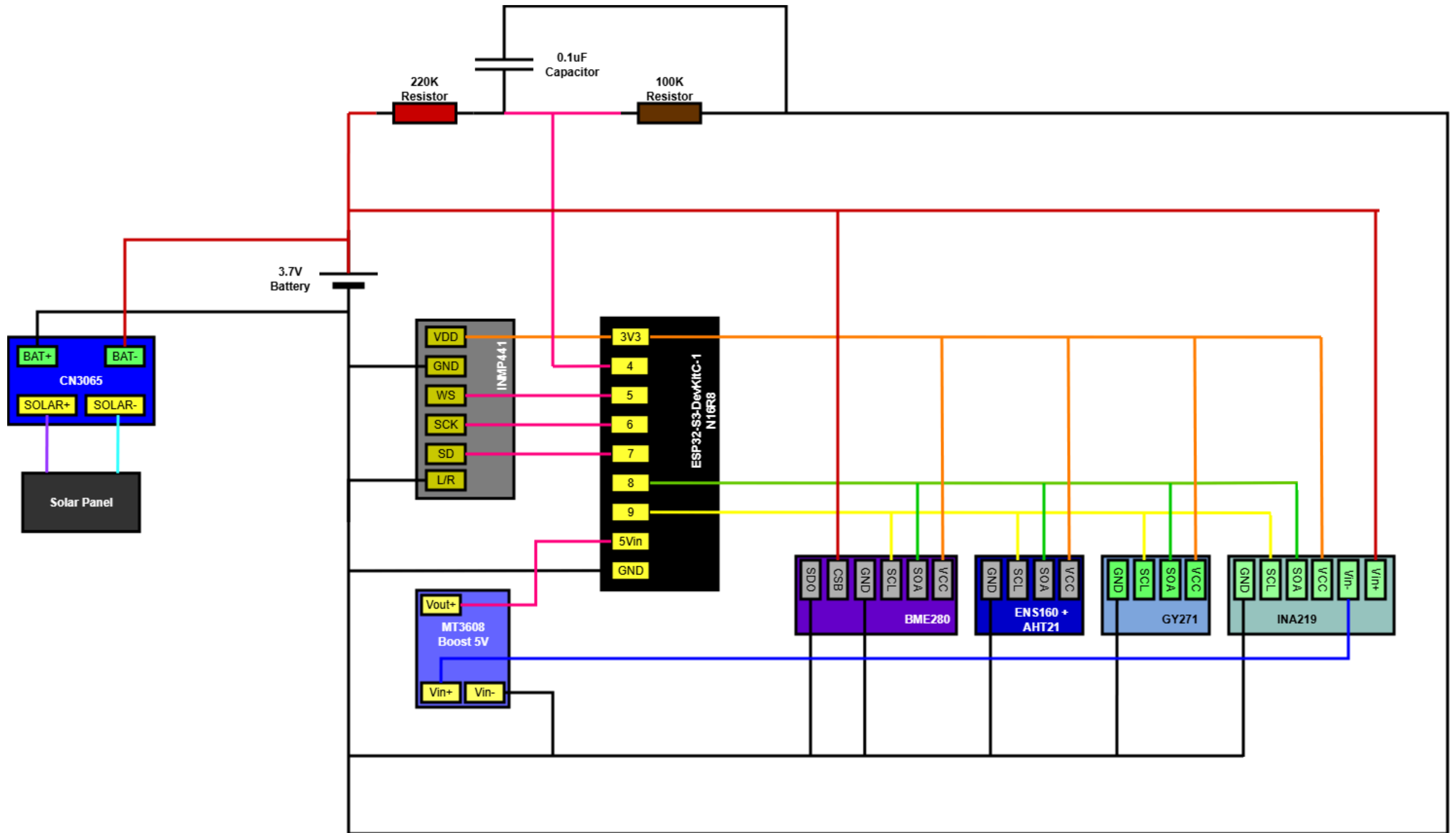
Brief Description

The component focuses on the Main Set sensor node. It uses an ESP32-S3 with environmental sensors, local storage, compression, BLE-triggered communication, and solar-based power support.

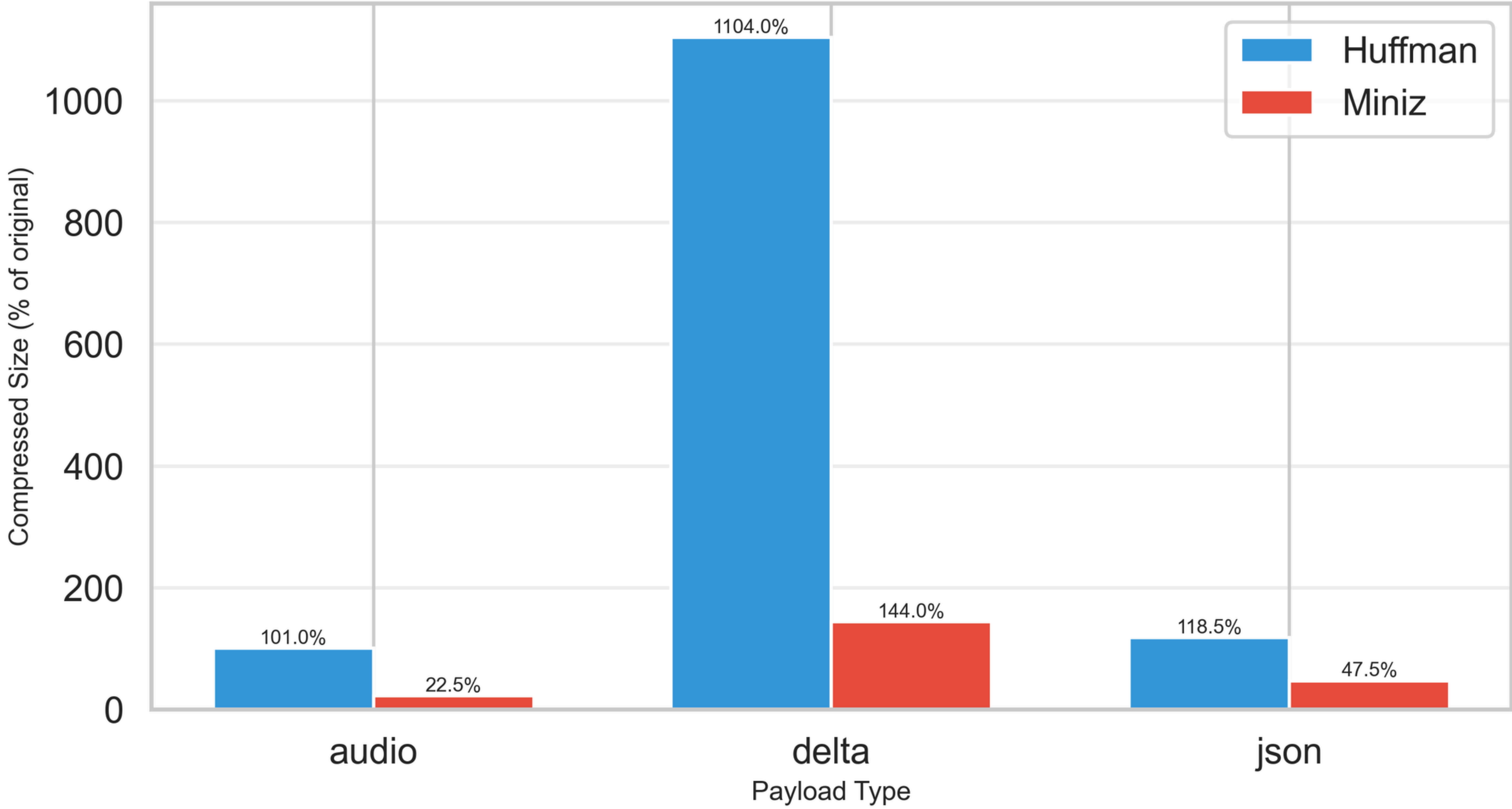
Result

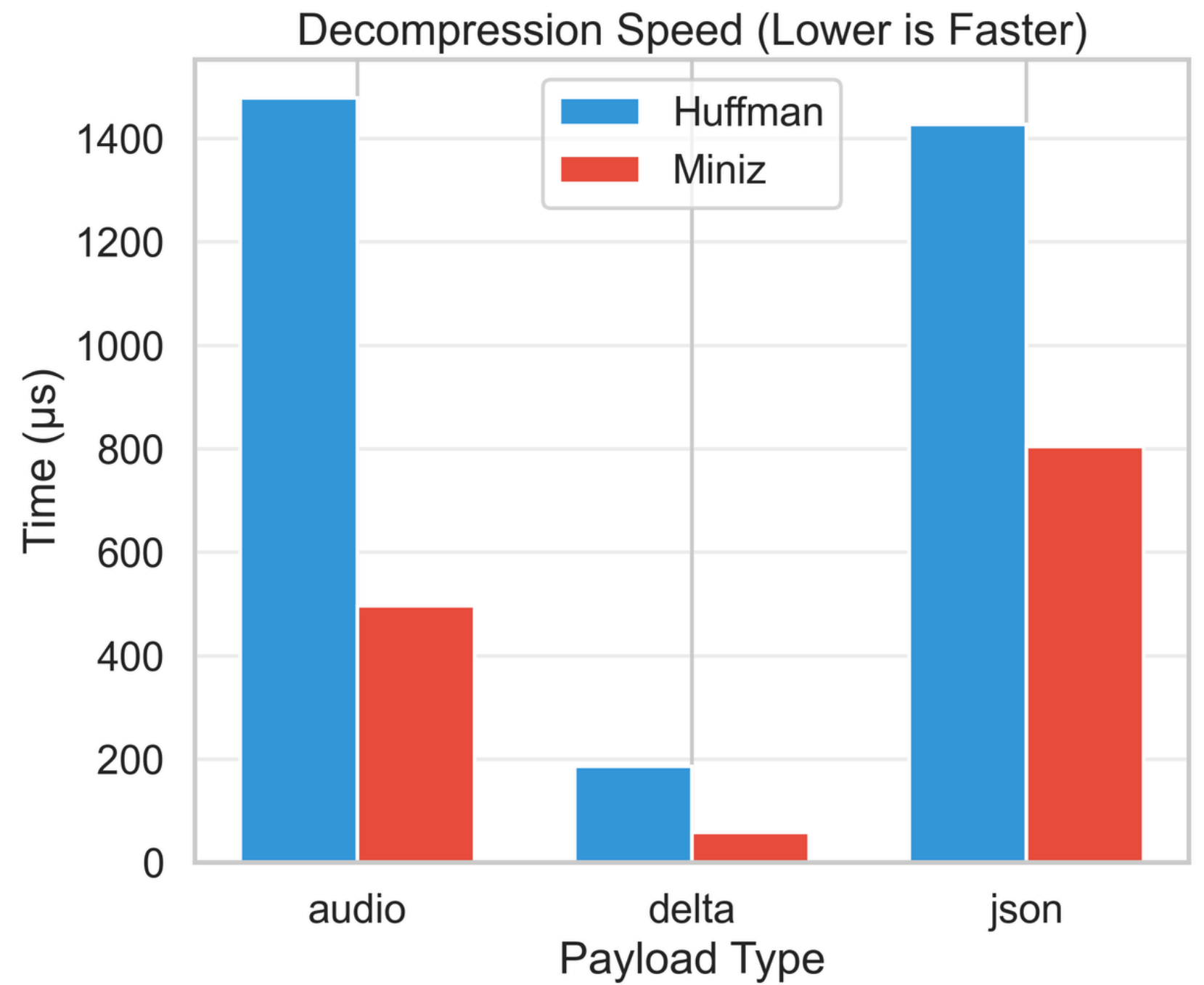
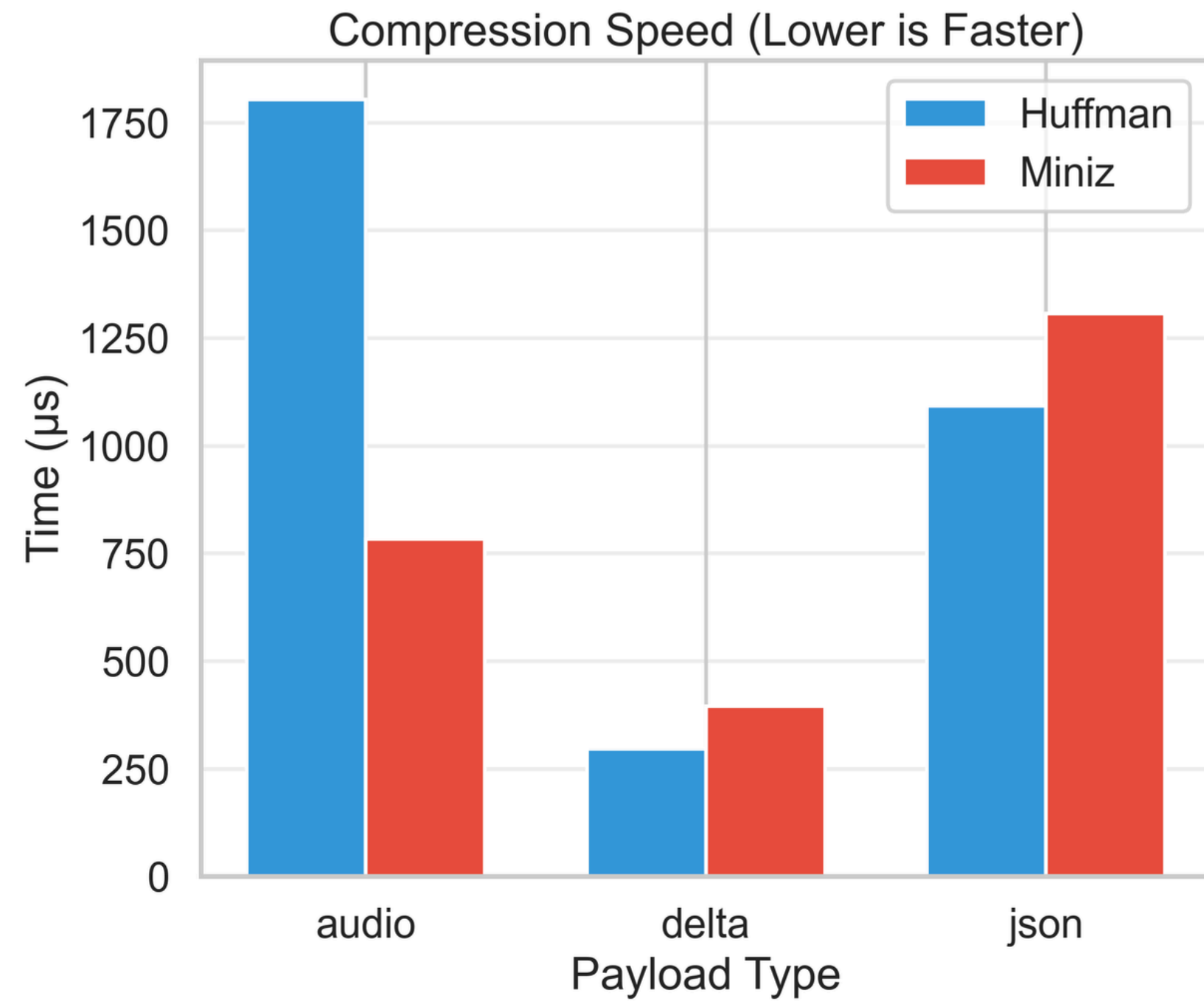
The node supports long-term autonomous operation, and miniz achieved better compression performance for structured sensor data.





Compression Efficiency: Huffman vs Miniz (Lower is Better)





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Cluster Formation & CH Election



Individual Component

STELLAR Cluster Head Election Algorithm

Brief Description

This component selects the best Cluster Head using multiple metrics and their variance instead of using fixed weighted metrics.

Metrics Used

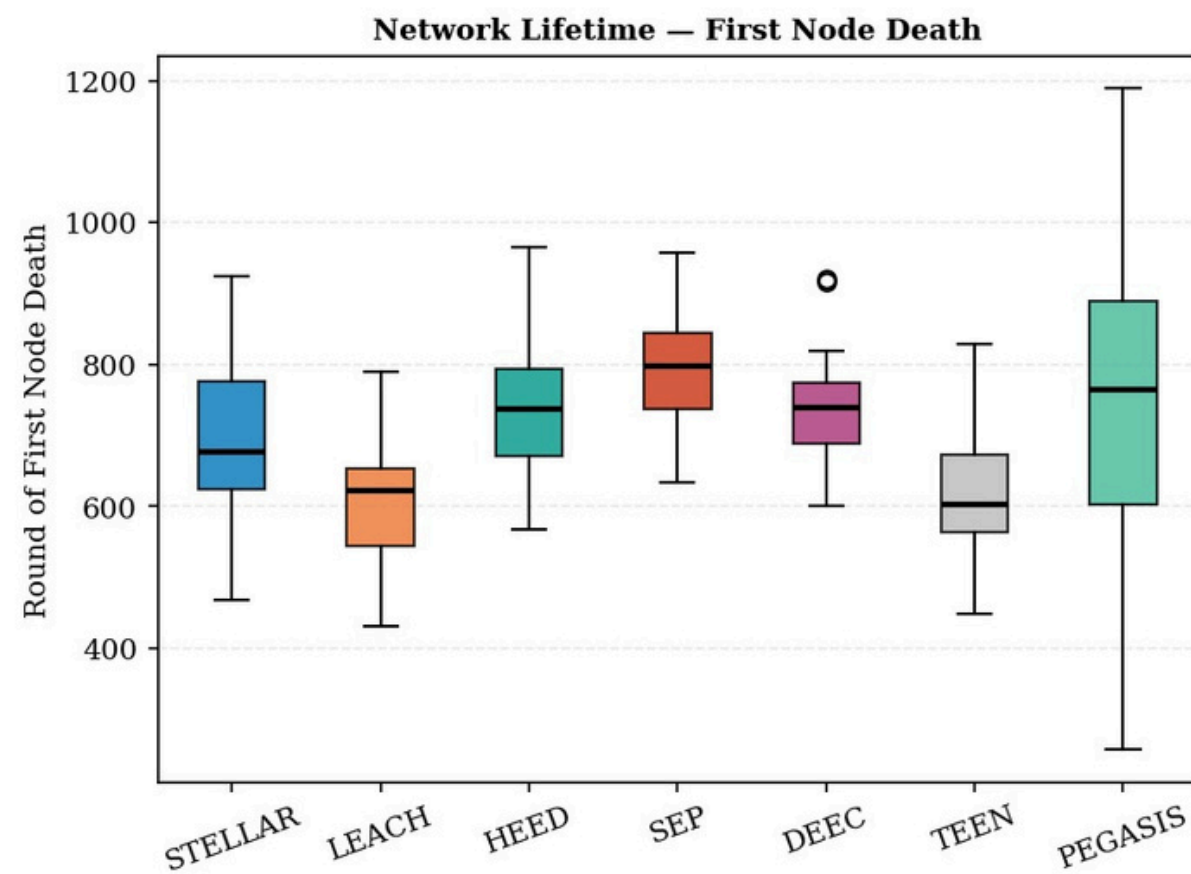
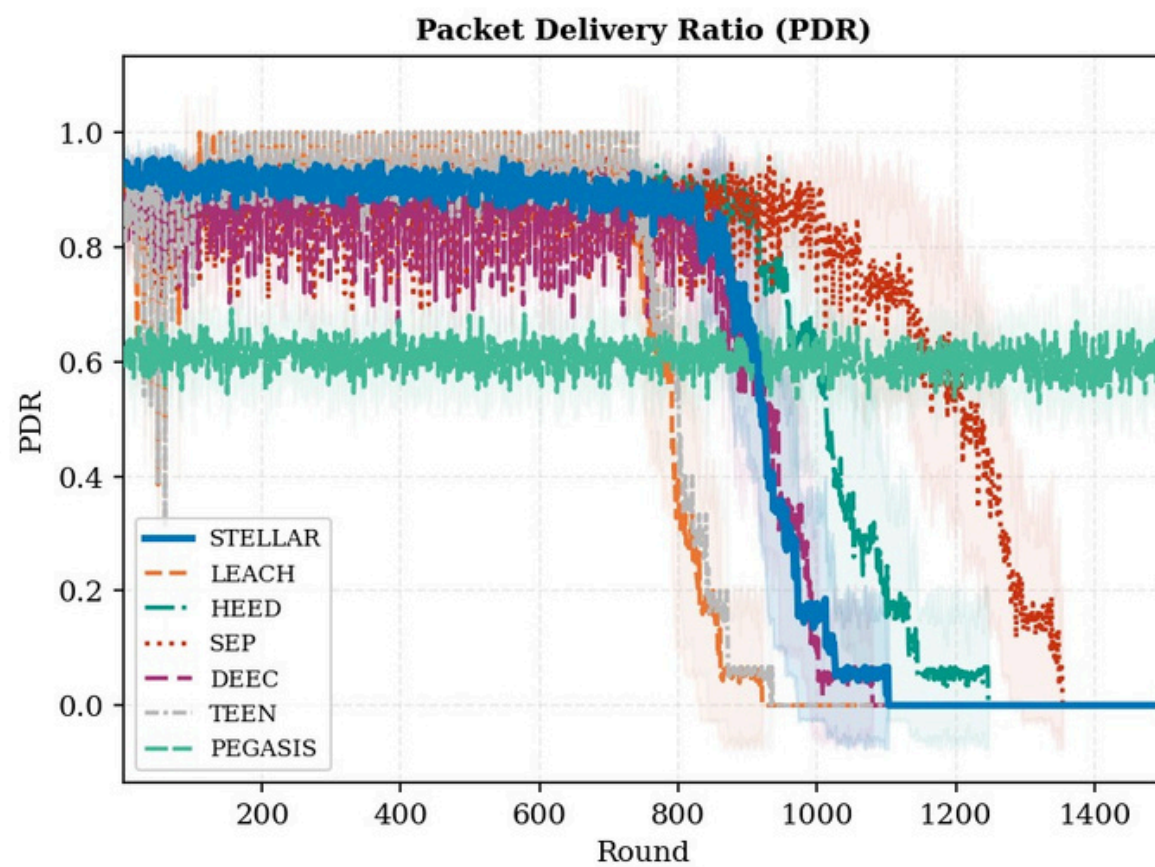
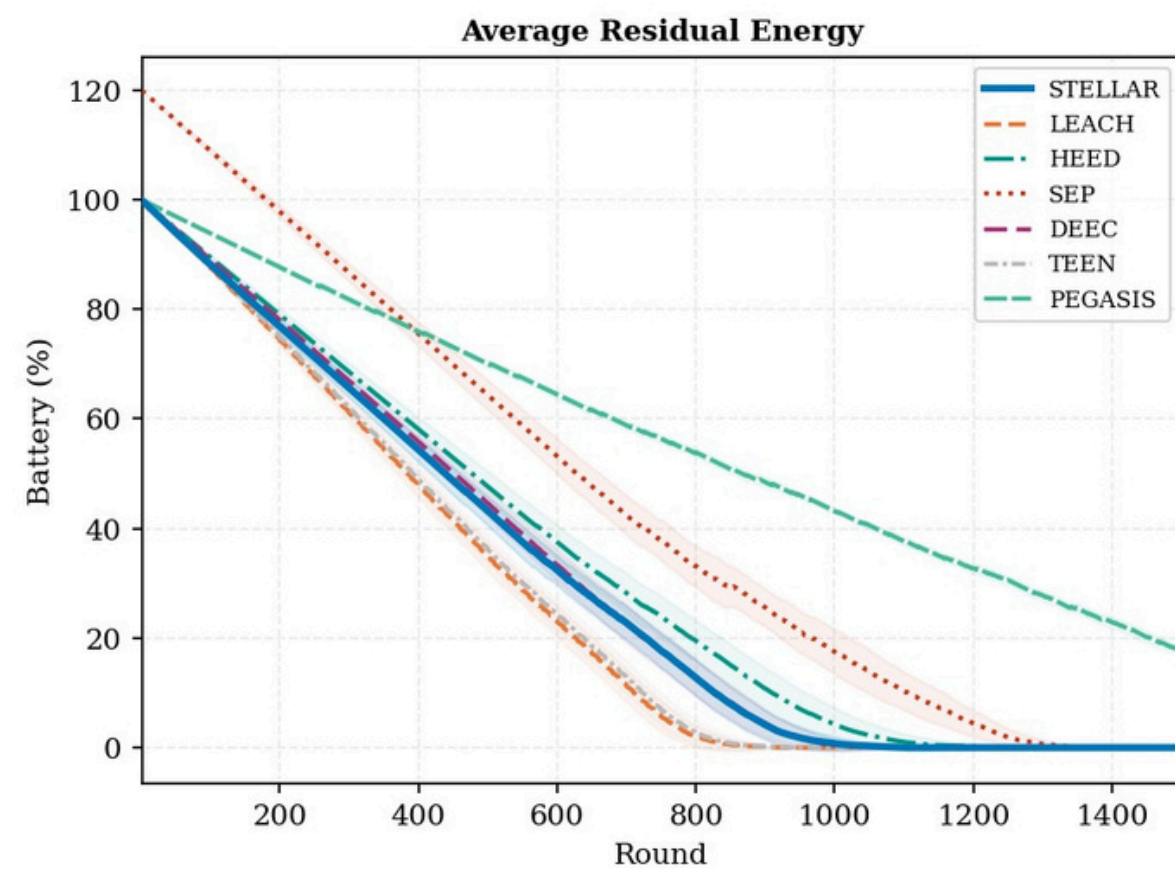
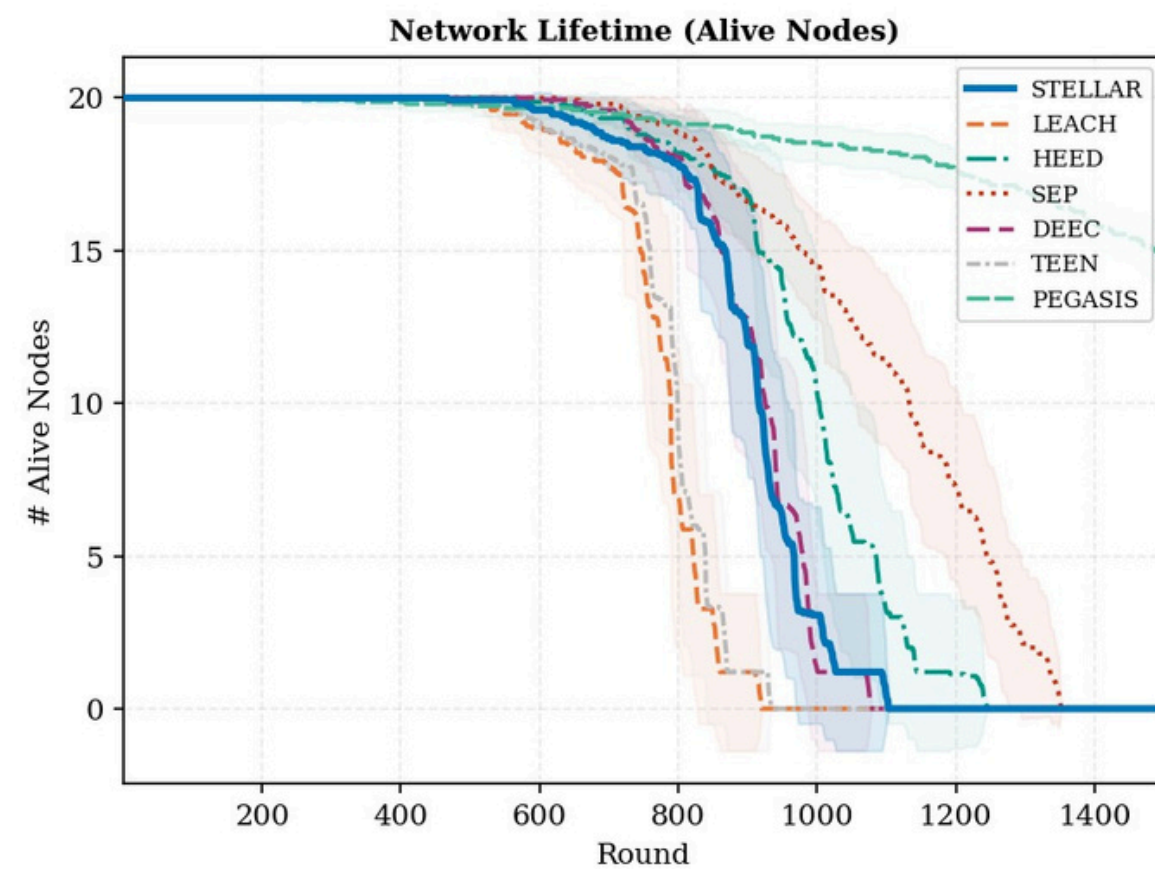
- Battery level
- Link quality
- Uptime
- Trust score

Key Technologies

- BLE advertisements
- HMAC-SHA256 authentication
- Lyapunov-stable adaptive weights
- Pareto filtering and Nash-product selection

Result

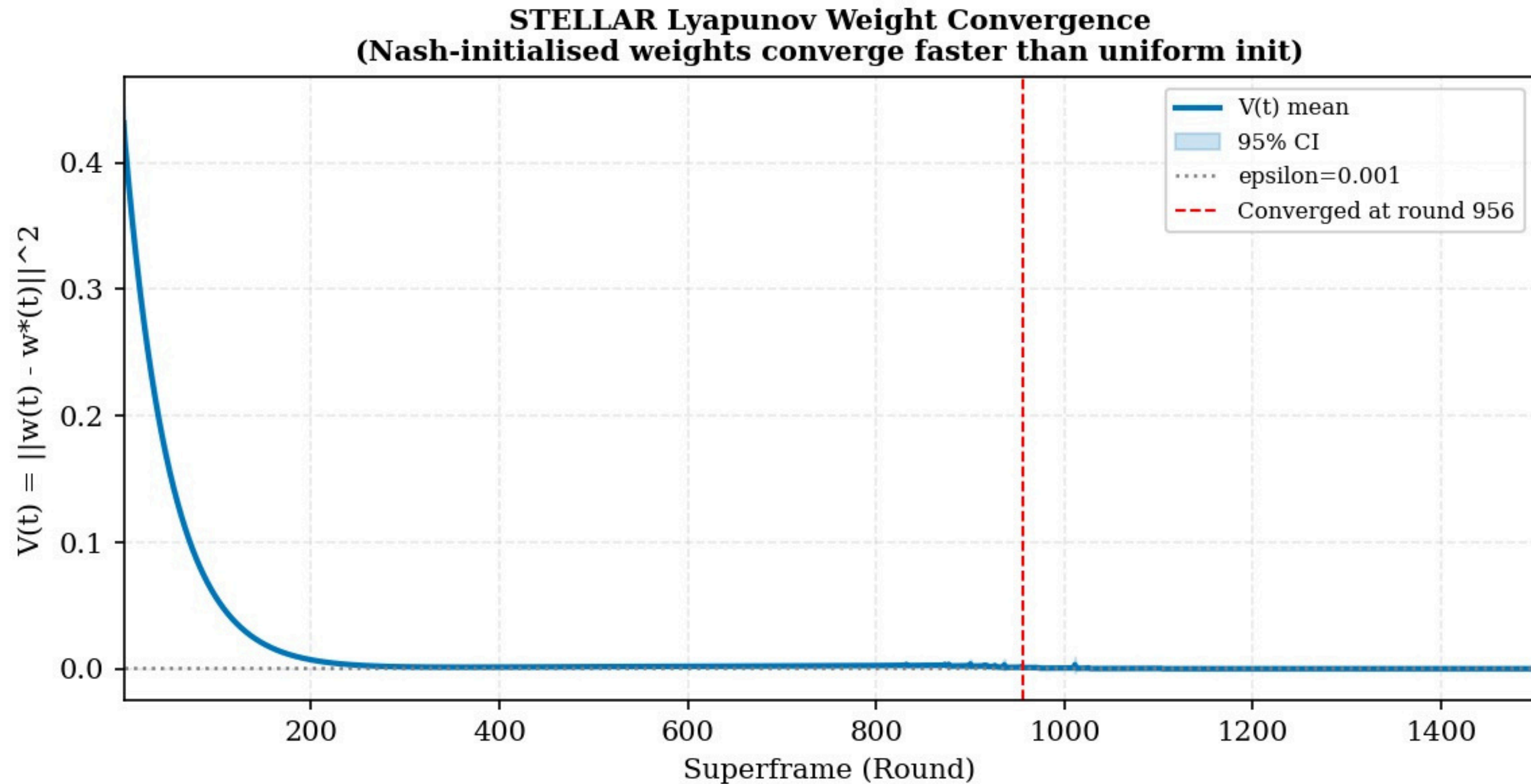
STELLAR achieved 91.8% mean steady-state PDR and showed better trust-aware CH selection compared with basic CH election methods.



Algorithm Simulation STELLAR vs 6 Baselines

Lyapunov Stability Guarantee

Convergence Graph (30 MC runs, 95% CI)



Statistical Significance

STELLAR vs 6 Baselines

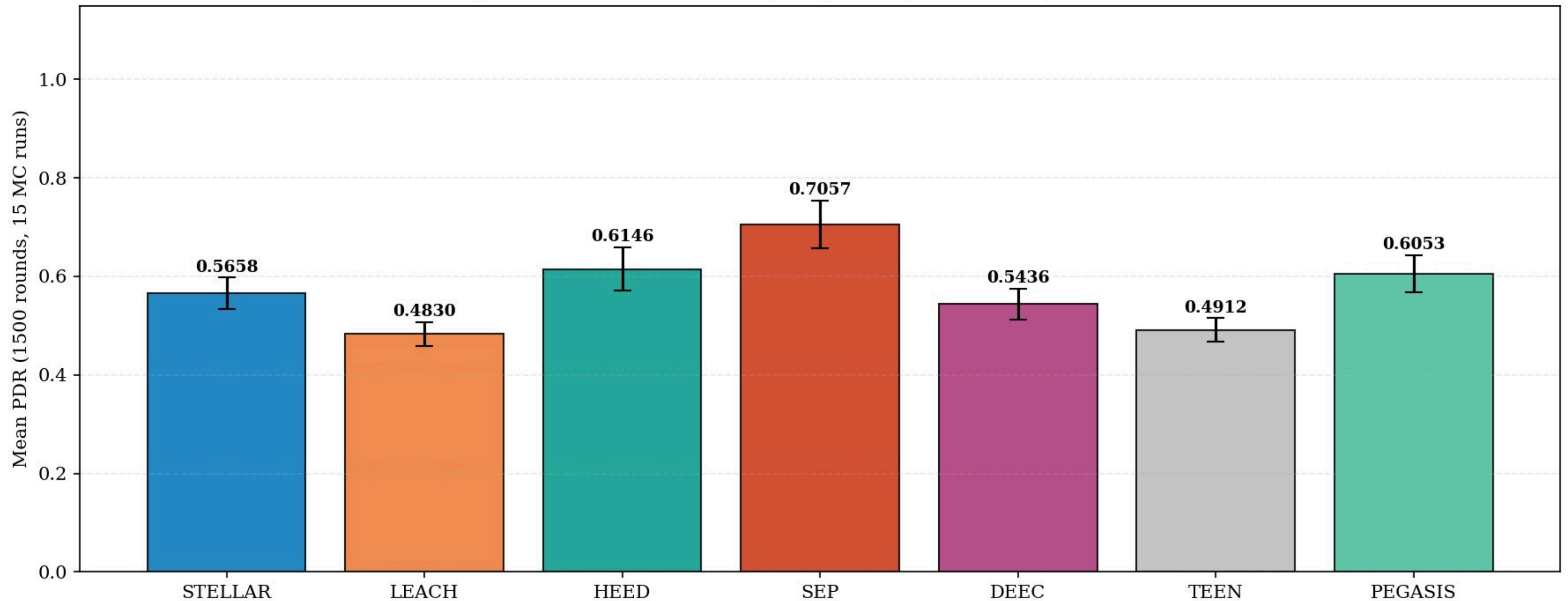
Table I — STELLAR Statistical Significance vs 6 Baselines
Complete STELLAR: Nash Bargaining + Pareto Filter + HMAC Trust + Lyapunov

Comparison	PDR t-test p	PDR MWU p	PDR Sig.(alpha=0.05)	Lifetime t p	Lifetime MWU p	Lifetime Sig.
STELLAR vs LEACH	0.000000	0.000007	YES	0.000011	0.000081	YES
STELLAR vs HEED	0.002248	0.003691	YES	0.008681	0.008972	YES
STELLAR vs SEP	0.000000	0.000011	YES	0.000001	0.000048	YES
STELLAR vs DEEC	0.076385	0.114987	NO	0.727317	0.618670	NO
STELLAR vs TEEN	0.000000	0.000009	YES	0.000062	0.000576	YES
STELLAR vs PEGASIS	0.005858	0.010122	YES	0.000000	0.000003	YES

Packet Delivery Ratio - All Algorithms

Mean PDR across steady-state rounds | 15 MC runs | Error bars = 95% CI

Fig 3 — Mean PDR: STELLAR (Complete Implementation) vs Baselines



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Cluster Data Communication



Individual Component

Energy-Efficient and Fault-Tolerant Cluster Data Communication

Key Features

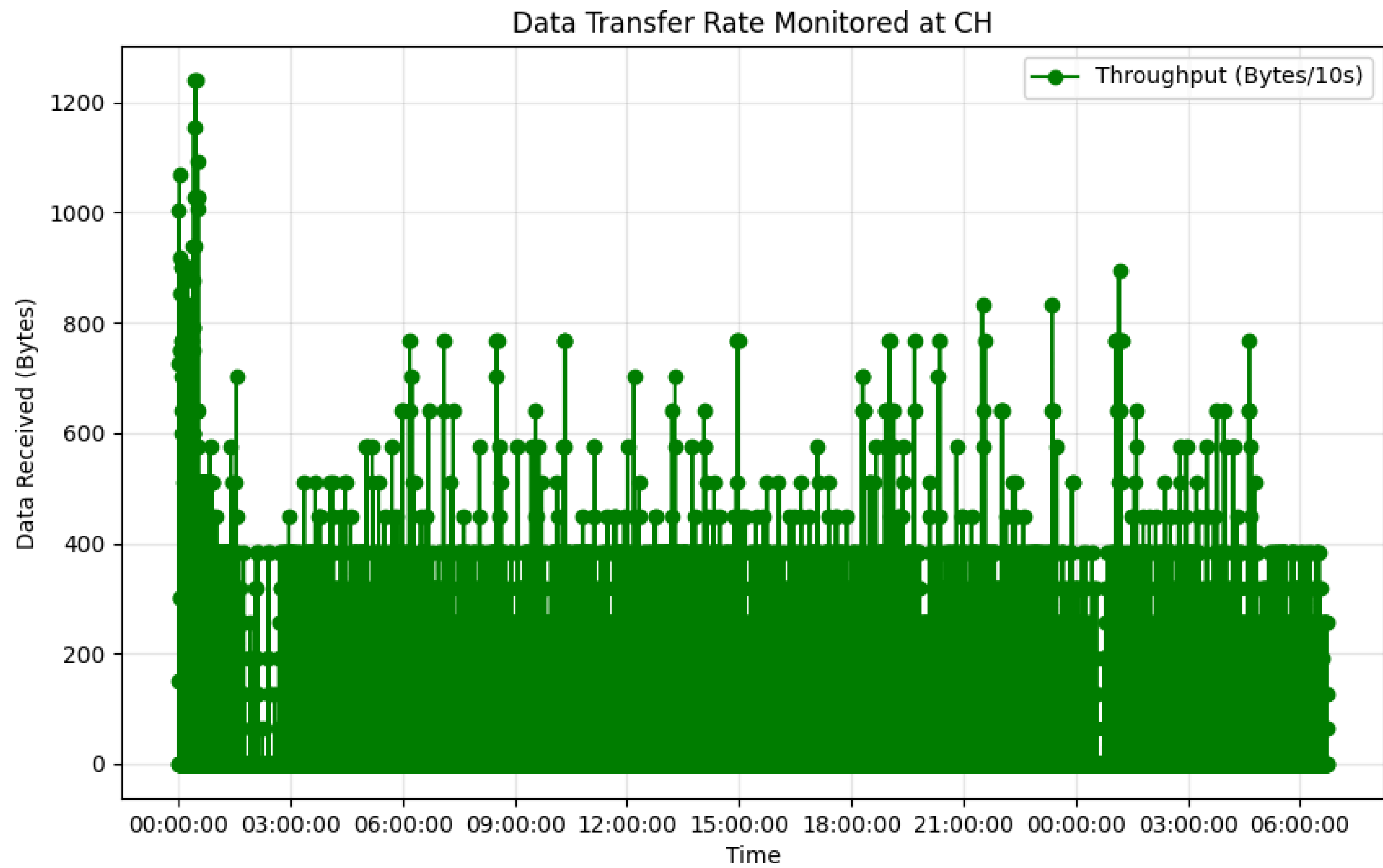
- Each MS gets a 10-second TDMA slot
- ESP-NOW burst transmission
- SPIFFS FIFO storage at CH
- MSLG binary chunk format
- CRC16 and CRC32 integrity checks

Brief Description

This component handles MS-to-CH communication using TDMA-based ESP-NOW scheduling and persistent CH-side storage.

Result

Testing showed zero TDMA collisions, around 80% node energy reduction, 100% SPIFFS recovery, and 65.2% end-to-end reliability in the initial firmware.



TDMA Superframe Parameters

Parameter	Value	Notes
Superframe period	40 s	STELLAR (20 s) + DATA (20 s)
STELLAR phase	20 s	BLE on — neighbour discovery & election
DATA phase	20 s	ESP-NOW only, BLE off
Phase guard (epoch offset)	5 s	Dead time before first slot; leaves 15 s for data
BLE pre-guard	3 s	Radio switched to Wi-Fi ch. before DATA boundary
Slot end margin	500 ms	Unsent chunks re-queued before slot cutoff
Election window	10 s	STELLAR score exchange; winner = highest Ψ score

TDMA Slot Parameters

Parameter	Value	Notes
Available slot time	15 s	DATA (20 s) – phase guard (5 s)
Slot duration (dynamic)	3 s / node	15 s ÷ 5 members at full cluster
Minimum slot	2 s	Members capped before slots shrink below this
User-requested slot	10 s / node	SLOT_DURATION_SEC — used for schedule validity window
Max cluster size	5 nodes	Split warning triggered above this
Max neighbours tracked	10 nodes	MAX_NEIGHBORS — neighbour table capacity
Schedule resend	every 5 s	CH re-broadcasts during DATA to catch late members

Superframe Timing Diagram



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UAV Routing & Data Mulling



Individual Component

A ROS 2 Node-Based UAV System for Autonomous Data Collection in Disconnected Wireless Sensor Networks

Brief Description

This component controls the UAV mission for collecting buffered data from Cluster Heads.

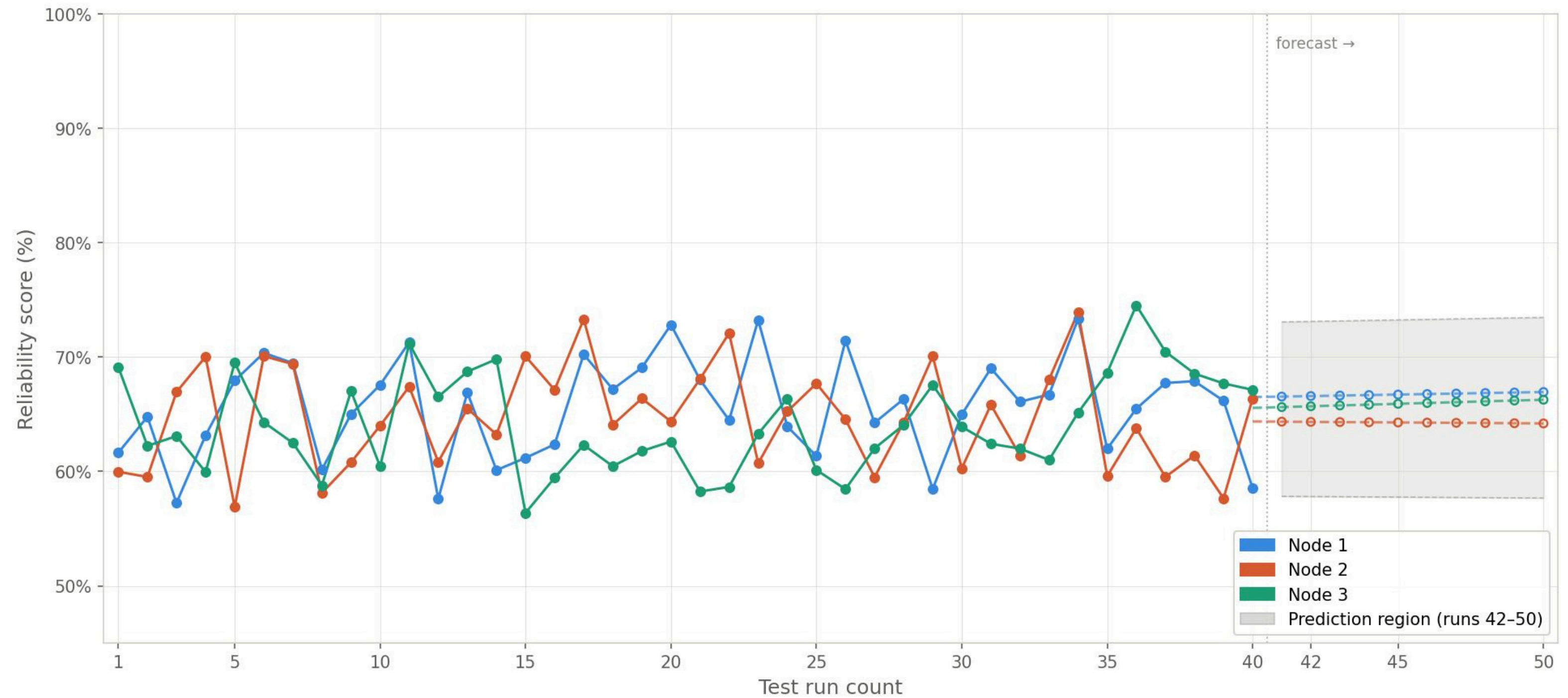
Result

Successfully demonstrated autonomous UAV-based data muling with reliable data collection from cluster heads and offloading at the base station.

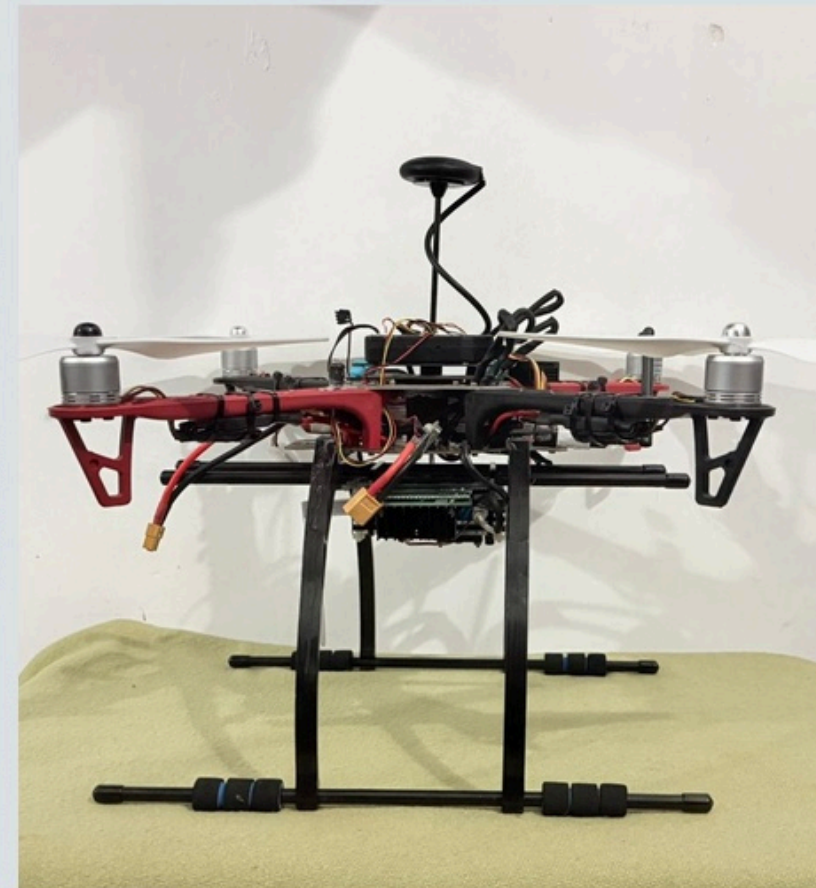
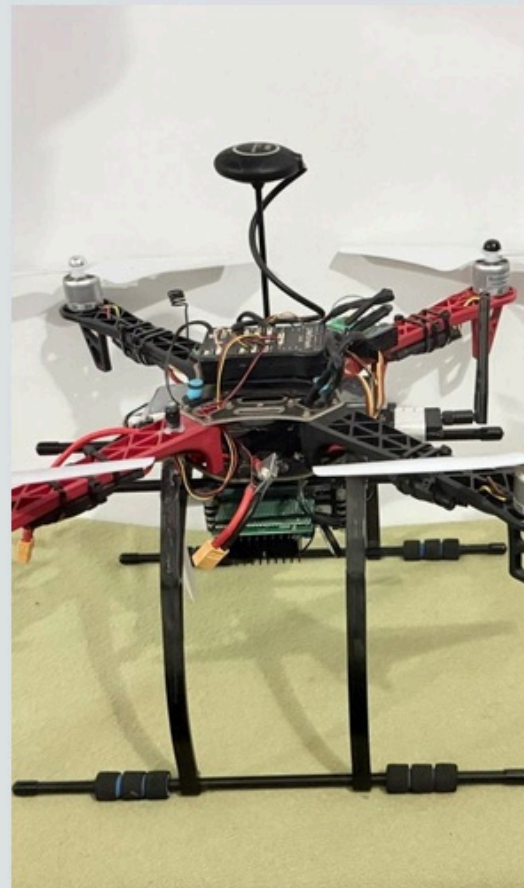
Key Features

- UAV as a mobile data ferry in disconnected sensor networks
- Raspberry Pi companion computer for onboard processing
- Pixhawk 2.4.8 flight controller running ArduPilot (Copter 4.6.3)
- MAVLink-based communication between flight controller and companion system
- ROS 2-based mission handling using modular nodes
- Waypoint-driven autonomous navigation with precise mission execution
- Altitude-aware hovering for reliable data collection from cluster heads
- Onboard data storage with batch offloading at the base station

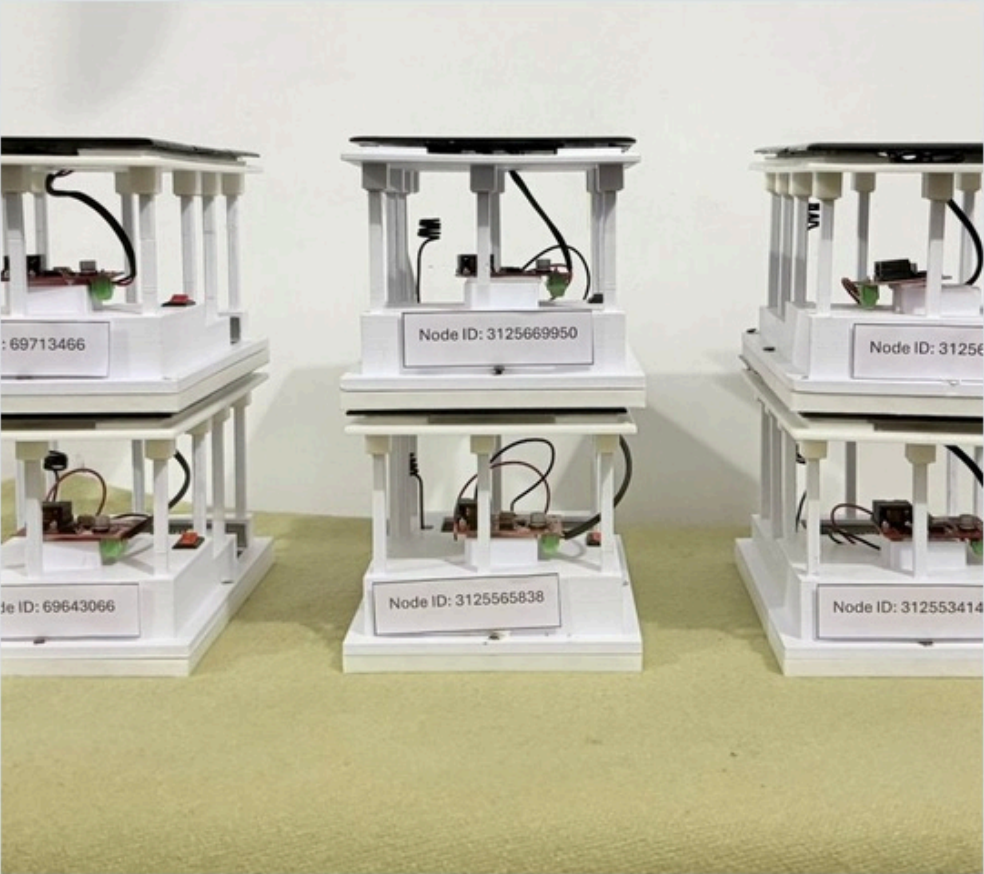
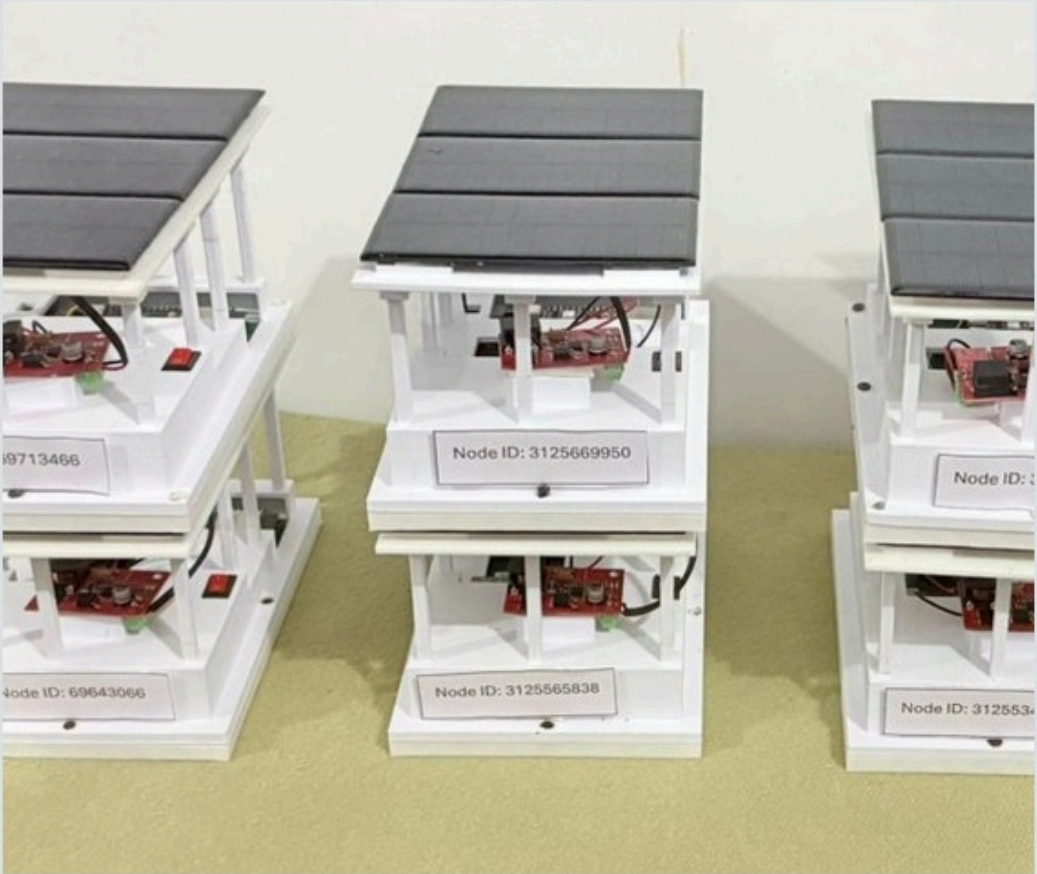
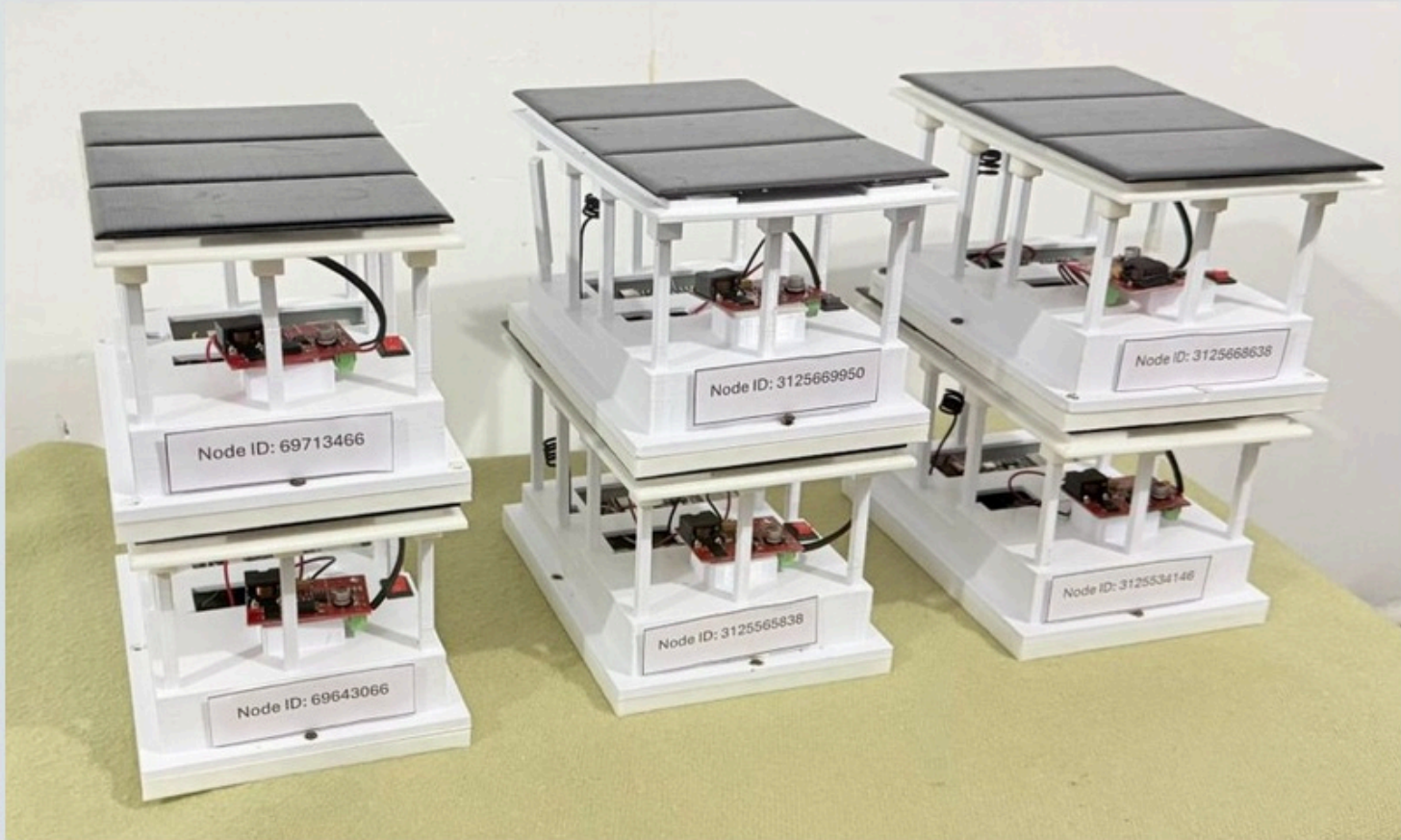
End-to-End Data Reliability upon each Arrival



UAV



Test Nodes



Results, Limitations & Commercialization

Main Results

- CH election convergence demonstrated
- TDMA communication stability demonstrated
- Persistent storage and recovery demonstrated
- UAV data ferrying pipeline demonstrated
- Dashboard update demonstrated
- End-to-end reliability: about 65% in initial firmware

Limitations

- End-to-end reliability needs improvement
- TDMA slot duration should be adaptive
- Larger outdoor testing is needed
- Full CH-to-CH handover can be improved

Future Improvements

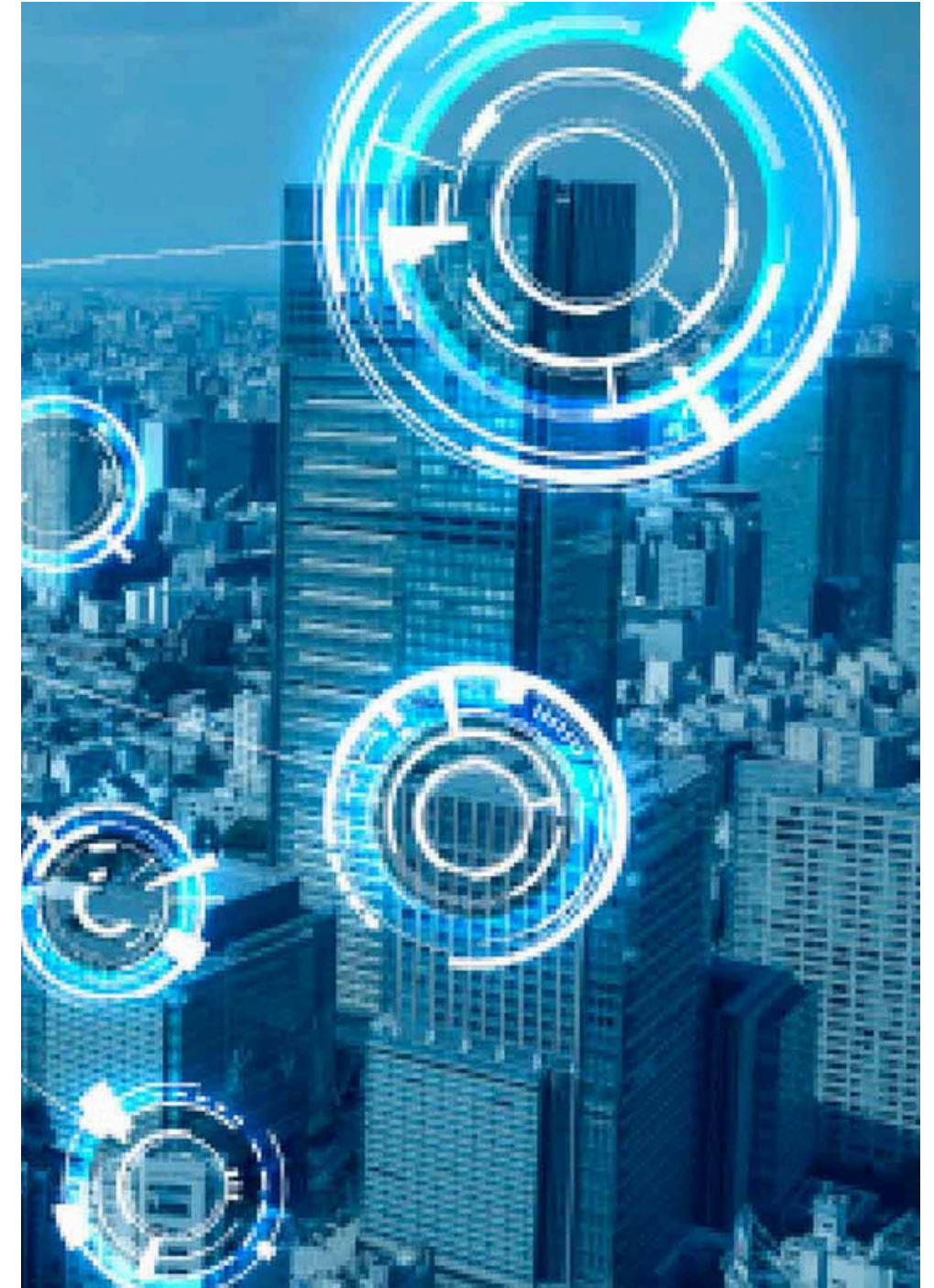
- Adaptive TDMA slot sizing
- Better UAV route optimization
- Larger field deployment
- Improved dashboard analytics

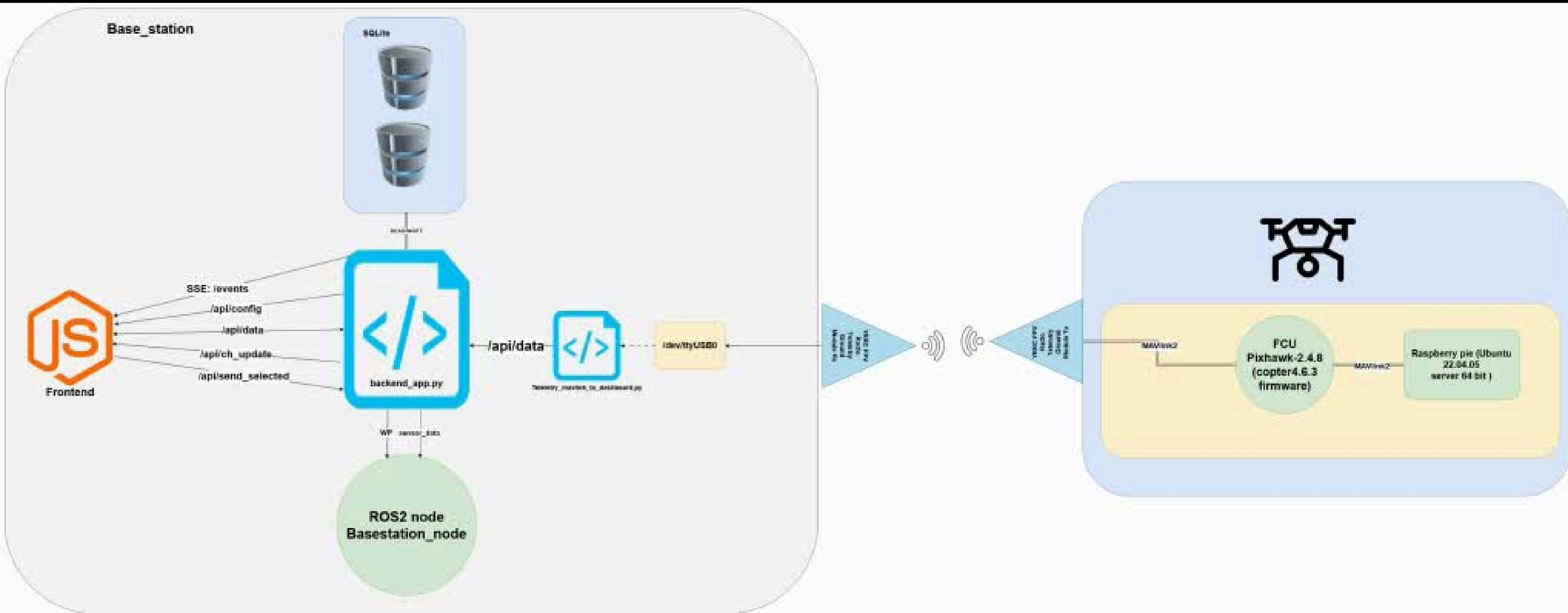
Commercial Potential

Possible use cases:

- Precision agriculture
- Wildlife monitoring
- Disaster response
- Border/environmental monitoring

Demo





WSN Demo



Thank You

